Full Semantic Constructed Network for Urban Use Classification From Very High-Resolution Optical Remote Sensing Imagery

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***Abstract*— Recently, semantic segmentation technology has been a research hotspot in optical remote sensing urban use classification. However, because of coupled semantic relations in very high-resolution and complex urban scenes, a more effective semantic description for pixelwise urban use interpretation has become a challenge. Then, aiming to set up a more effective semantic description, the effective receptive field (ERF) is analyzed in general convolutional neural networks. The unreasonable ERF distribution in the stacked convolutional layers of the encoder would lead to a large amound of small ERFs and fewer not large enough ERFs that form a naive semantic description in decoder. Therefore, in this article, a novel full semantic constructed network (FSCNet) is proposed to improve the naive semantic description and set up an effective semantic description. First, to avoid noise from shallow feature layers, a residual refinement convolution is designed to optimize the full-scale skip connections based on the U-shaped encoder–decoder. Second, an interscale fusion module is newly designed for multiscale feature fusion, which can generate three initial semantic modalities that are prepared for redefining the full semantic description. Third, a multiscale local context spatial attention module and boundary supervision are designed for an initial shallow semantic modality to capture the pure boundary information, and then, pyramid spatial pooling is employed for an initial deep semantic modality to further enlarge the ERF and obtain more abstract global information. Next, a self-calibration convolution combined with the atrous spatial pyramid pooling is designed to rectify and enrich an initial middle semantic modality, which can improve the naive semantic description and bridge the semantic gap between the redefined shallow and deep semantic modalities to advance the full semantic feature fusion. Finally, extensive experiments are carried out on three benchmarks (e.g., ISPRS Vaihingen, Potsdam, and DLRSD), and comparative results show that the proposed FSCNet can get remarkable performance compared to state-of-the-art (SOTA) methods. Besides, the code is available at https://github.com/DorisCV/FSCNet.**

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I. Introduction

SEMANTIC segmentation is an essential task of assigning the class label for per-pixel in a given image, which can be employed for automatic driving, urban planning, human pose estimation, and change detection [1], [2], [3], [4], [5], [6], [7] [8]. Recently, urban use classification from optical remote sensing imagery has become a hot research topic for semantic segmentation. However, with the development of very high resolution (VHR), many interior details are presented in complex urban scenes; consequently, establishing a more effective semantic description for complex and coupled semantic relations dealing with pixelwise semantic segmentation for urban scenes is a challenging task.

According to more effective semantic feature construction for accurate urban use classification, several works have been studied, and they can be divided into two categories: object-level and pixel-level-based methods. First, object-level-based methods [9], [10], [11], [12], [13] usually employ low-level feature descriptors (e.g., local binary patterns, superpixels, and bag-of-visual-words) for object-level descriptions, and then, condition random fields, Markov random fields, and graph cuts are often used for semantic relation modeling. However, due to their insufficient semantic feature generalization ability, these object-level-based methods often exhibit poor performance for pixelwise urban use classification. Second, thanks to the encoder–decoder of convolutional neural network (CNN) possessing pixelwise semantic description ability, the pixel-level-based methods of semantic segmentation networks (e.g., FCN [14], Deeplab-v3+ [15], SegNet [16], Unet [17], GCN [18], PSPNet [19], Unet++ [20], Unet3+ [21], and BiSeNet [22]) have been widely developed and migrated into the pixelwise urban use classification. For instance, based on the FCN [14] or BiSeNet [22] framework, Lin et al. [23] and Liu et al. [24] merged the global–local or high- and low-level features by parallel CNN feature extractors with multiscale inputs for urban use classification. Sang et al. [25], Lei et al. [26], and Peng et al. [27] enhanced high- and low-level features and then advanced feature fusion by designing

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a specific attention mechanism for refined urban used classification. Li et al. [28], Dong et al. [29], and Ding et al. [30] considered not only merging the high- and low-level features but also introducing the hierarchical feature expression into the final semantic feature representation, which can improve urban use classification performance. Then, in view of the one- or two-path encoder–decoder of Deeplab-v3+ [15], GCN [18], Unet [17], Unet++ [20], and Unet3+ [21] for semantic feature generalization, Chen et al. [31], Mou and Zh [32], and Li et al. [33] all adopted skip connections to advance different level feature interactions; thus, these studies focused on optimizing the feature representation of the encoder and the decoder, and then fused features with multiscale effective receptive fields (ERFs) for urban use interpretation. Although many efforts have been made for urban use classification, owing to the scattered distribution of entities and complicated semantic relations in VHR and complex urban scenes, the abovementioned studies cannot provide fine-gained pixelwise urban use classification results, which are mainly caused by the limited semantic feature representation of the encoder–decoder architecture and suboptimal skip connection strategy for feature fusion.

Recently, some studies have begun to improve the semantic feature generalization ability to facilitate refined pixelwise urban use classification. Sang et al. [39] and Chen et al. [40] considered employing self-attention and channel attention (CA) combined with a residual structure to redesign encoder–decoder architecture while simultaneously improving pixelwise classification and localization ability; thus, they can obtain quite good urban use classification results. Then, related to the essence of semantic feature description, Chen et al. [41] and Keiller et al. [42] studied the importance of ERFs to capture valid multicontext for semantic feature generalization, and both evidently proved that combining ERFs to mix multiple valid multicontext information can construct a more valid semantic description for pixelwise interpretation. In addition, Zhou et al. [43], Mou et al. [44], and Deng et al. [45] considered improving the semantic feature generalization ability from class-guided or spatial relation matters. However, only the classification ability or localization capability is considered to leverage the semantic feature generalization ability and may have limitations for refined pixelwise urban use classification; therefore, the classification and localization abilities for each pixel are important. Next, some studies [46], [47], [48], [49], [50] commenced rethinking semantic feature generalization to redefine the semantic feature as two parts, including the semantic boundary and the main body. Therefore, they introduced the boundary-aware feature extraction module into the encoder–decoder, while multitask supervision was utilized to simultaneously optimize boundary localization and pixelwise classification. Nevertheless, related to boundary-aware feature extraction in stacked convolution layers, consecutive convolution structures would lose much localization information to generate large ERFs in deep feature layers, so they cannot provide precise semantic boundary prediction. Otherwise, when these small ERFs in the shallow feature layer are utilized to predict the semantic boundary, they introduce considerable noise that affects semantic boundary prediction because of scanty ERFs. In addition, whether for the one- or two-path encoder–decoder architecture, although shallow and deep features can individually capture the information of the semantic boundary and the main body, the great opposite function of shallow and deep features for pixelwise classification and localization is usually called a severe semantic gap. If these shallow and deep features are blindly fused for semantic feature generalization, it is conceivable that the severe semantic gap would worsen the generalized ability for pixelwise urban use classification. In general, to further improve refined urban use classification performance, how to redefine semantic feature generalization is a worthy issue to be continuously explored.

In this article, to explore a more effective semantic feature generalization method for refined urban use classification from VHR optical remote sensing imagery, our study focused on the essence of semantic feature descriptions and analyzes the ERFs in general CNNs. Subsequently, we find that there is an unreasonable ERF distribution in stacked convolution layers of the encoder, which only contains a large number of small-scale ERFs and fewer ERFs that are not large enough to form the semantic feature description, as shown in Fig. 1. Here, although hierarchical feature fusion and multiscale feature fusion methods are extensively studied by existing methods, they both ignore this unreasonable ERF distribution, which severely limits the upper bound of performance. Then, in view of a large number of small-scale ERFs with insufficient receptive fields, regardless of whether the valid or invalid boundary or texture information is randomly captured for pixelwise localization, this invalid boundary and texture information can be considered as noise that worsens the semantic feature description. In addition, the fewer ERFs that are not large enough also cannot capture effective and global abstract information for diverse pixelwise classification. Therefore, only containing a large number of small ERFs and fewer ERFs that are not large enough would capture the limited variety of context information, generate a naive semantic description with considerable noise information at the decoder, and it is difficult to adapt to complex urban scene interpretation. Thus, a novel full semantic constructed network called (FSCNet) is proposed to set up a more effective semantic feature description. Here, in order to improve the naive semantic description, which makes it difficult to process VHR complex urban scenes, the shallow, middle, and deep semantic modalities are redefined to capture their specific semantic information. First, related to a large amount of noise from shallow feature layers, a residual refinement convolution (RRC) is designed to optimize the full-scale skip connections for the U-shaped encoder–decoder, and it can avoid the noise from shallow feature layers being introduced into the final semantic feature description. Second, to set up three initial semantic modalities from hierarchical features, an interscale fusion (ISF) module is designed to fuse multiscale features to generate three initial semantic modalities that can gradually fuse several adjacent scale features and avoid interference between different scale features. Third, based on the initialized three semantic modalities, a multiscale local context spatial attention (MLCSA) module and boundary supervision are

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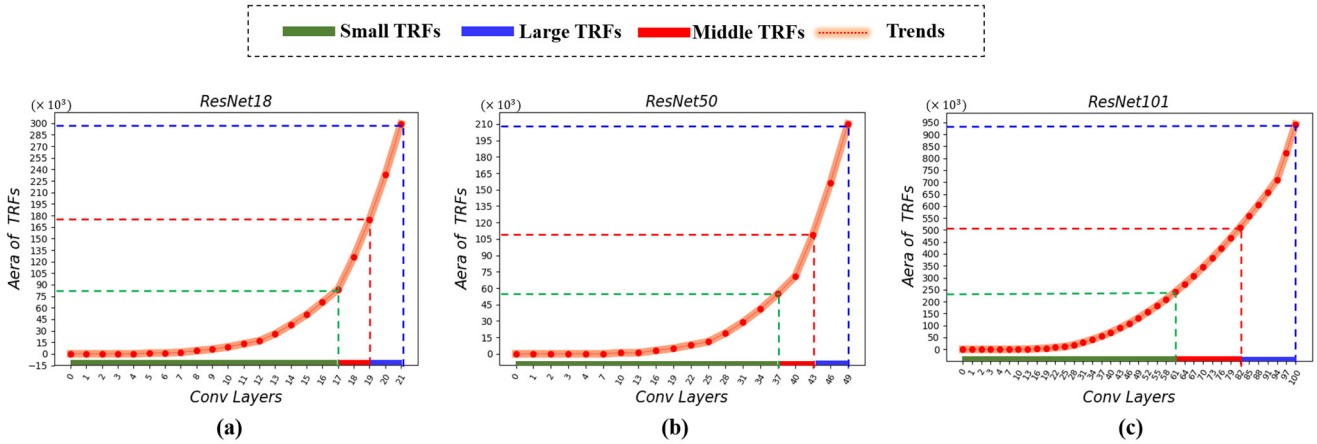


Fig. 1. Example diagram of TRFs changes with convolution layer going deeper; red dots are the discrete sampling values of the area of TRFs in each layer; green, red, and blue color lines represent small, middle, and large TRFs. (a) Changes in area of TRFs in Resnet18. (b) Changes in area of TRFs in Resnet50. (c) Changes in area of TRFs in Resnet101.

designed for the initial shallow semantic modality to learn the pure boundary information, and then, pyramid spatial pooling (PSP) is utilized for the initial deep semantic modality to further enlarge ERFs to capture more abstract global information. Then, due to the severe semantic gap between the pure boundary of shallow semantic modality and abstract global information of deep semantic modality, a novel self-calibration convolution (SCC) combined with atrous spatial pyramid pooling (ASPP) is designed to rectify and enrich an initial middle semantic modality, which can bridge a semantic gap between redefined shallow and deep semantic modalities, and improve the naive semantic description for better fused semantic feature description. Finally, extensive experiments are performed on three benchmarks (e.g., ISPRS Vaihingen, ISPRS Potsdam [51], and DLRSD [52]). Then, the results indicate that the proposed FSCNet outperforms the other state-of-the-art (SOTA) methods for refined pixelwise urban use classification. In general, the contributions of this study are summarized in the following.

1) Unlike general multiscale feature fusion in CNN-based encoder–decoder, a novel full semantic description strategy is proposed by analyzing the unreasonable ERF changes in the general encoder and then redefining the semantic modalities to improve the semantic description ability and reasonably bridge the severe semantic gap. In addition, through our designed full semantic construction method, the proposed FSCNet can obtain accurate urban use classification results even with complex pixelwise semantic relations (i.e., the 17 urban use classification classes on the DLRSD dataset).

2) An ISF module is designed to adaptively integrate these adjacent different scale features from the optimized full-scale skip connections, and it can also avoid complicated interscale interferences caused by straightforward concatenated or added operations for different-scale semantic feature fusion and make interscale feature integration more reasonable.

3) An MLCSA module is proposed to optimize the shallow semantic modality to get more effective boundary information; in addition, combined with a multitask supervision strategy, the designed MLSCA module can not only remove noise in the shallow semantic modality to optimize the rebuilt full semantic description but can also provide refined boundary prediction from complex optical remote sensing urban scenes.

4) An SCC combined with ASPP is designed to rectify and enrich the classification and localization ability of the middle semantic modality, which can effectively bridge the semantic gap between shallow and deep semantic modalities and endow the fused semantic feature with the ability to simultaneously juggle the pixelwise localization and classification ability.

II. Related Work

This article aims to construct a more effective semantic description for refined pixelwise urban use classification. Thus, in this section, we briefly review related studies and clarify our study motivation.

A. Refined Pixelwise Urban Use Classification

With the development of VHR optical remote sensing, refined urban use classification is urgently required in practical applications; therefore, several studies have focused on setting up a model to provide refined pixelwise classification results. Related to the CNN-based encoder–decoder architecture, some studies [23], [24], [25], [26], [27], [28], [29], [30] have designed different feature fusion methods to generate a better semantic description from the encoder for urban scene interpretation. Then, Chen et al. [31], Mou and Zhu [32], Li et al. [33], Treerapong et al. [34], Cheng et al. [35], Liu et al. [36], [37], Foivos et al. [38], Sang et al. [39], Chen et al. [40], [41], and Keiller et al. [42] mostly focused on optimizing different scales of encoding and decoding feature representation and fused multiscale features with varied ERFs to model the semantic feature description for complex urban use classification. In addition, Zhou et al. [43], Mou et al. [44], and Deng et al. [45] employed the attention mechanism to facilitate the encoder–decoder architecture to capture better pixelwise spatial and categorical information for urban use classification. Subsequently, studies [46], [47], [48], [49], [50], and [55] introduced boundary awareness to improve pixelwise

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urban use classification performance, and these studies both redefined the semantic feature description into two parts (i.e., the semantic boundary and main body) to better balance the classification and localization ability for pixelwise prediction. From the abovementioned studies, we find that, in recent years, refined pixelwise urban use classification has become a hot research topic; thus, in this article, we also pay attention to establishing a more effective semantic feature description to improve the refined pixelwise urban use classification performance.

B. Semantic Feature Description

According to CNN-based encoder–decoder architectures for semantic feature generalization, the existing methods can be mainly classified into two categories: single- and two-path semantic feature description strategies. Here, the classical FCN [14] is a standard single-path semantic feature description method; however, it only utilizes the deep feature layer as the final semantic feature description for pixelwise prediction, which results in poor localization performance. Then, the Deeplabv3+ [15] family is the widely used single-path method for semantic feature descriptions, and due to the ASPP applied for a relatively high-resolution feature map rectifying the ERFs to capture various context information, it can provide better pixelwise prediction results compared to FCN [14]. In addition, following the single-path method, BiSeNet [22] adopts different convolution kernels to individually construct two independent branches, which can make the model ensure the classification and localization ability individually and then fuse two branches' features for semantic feature description. Except for the single-path semantic feature description method, the U-shaped SegNet [16] and UNet [17] are representative two-path methods, which are also widely utilized for dense pixelwise prediction tasks. Then, because of the naive skip connection method of SegNet [16] and Unet [17], Unet++ [20] and Unet3+ [21] are developed to improve the feature fusion by abundant skip connections, and deep feature supervision is designed to ensure the fusion quality of intermediate features. However, although semantic feature description strategies are widely studied, for refined urban use classification tasks, when ASPP is applied in a single-path way, large atrous rates would result in a gird effect that degenerates the final semantic feature description and leads to poor performance. Thus, many studies [24], [30], [37], [41], [42], [45] are dedicated to improving the single-path method of Deeplabv3+ [21] to obtain better refined urban scene interpretation. Subsequently, although the multiscale feature fusion of the U-shaped two-path method is also widely considered for semantic feature descriptions and achieving refined urban use classification, many studies [27], [31], [32], [33], [34], [38], [40], [46] have indicated that the noise from shallow feature layers should be mitigated or avoided for final semantic feature fusion. Therefore, for both single-path and U-shaped two-path semantic feature descriptions, it is difficult to balance the abilities of pixelwise classification and localization predictions, which usually yields a poor performance for refined urban use classification. In this article, we focus on studying the U-shaped two-path method to further improve the ability of semantic feature description and achieve refined pixelwise prediction in VHR complex urban scenes.

C. MultiTask Supervision Strategy

The multitask supervision strategy is widely used for deep learning models, and existing methods [46], [47], [48], [49], [50], [53], [54], [55] have been proven to improve the robustness and generalization ability of deep learning models. For instance, Dai et al. [53] introduced semantic segmentation supervision into object detection to simultaneously achieve instance segmentation and improve detection performance. Bischke et al. [54] also designed a multitask loss, including the truncated distance of boundary and semantic segmentation terms, to jointly facilitate pixelwise prediction for building footprint classification. Wang et al. [48] designed boundary-aware multitask learning to introduce multitask losses, including semantic segmentation, height, and boundary predictions into Deeplabv3+ [15] to get refined predictions. Li et al. [49] introduced boundary supervision to supervise the shallow feature optimization and promote land-use classification performance. Furthermore, in our previous study [25], we mainly focused on the significant differences among multiscale features in encoder–decoder to rectify fallible results (i.e., the boundary and various interior textures). In general, all these studies demonstrate that suitable multitask supervision can make the built model obtain refined pixelwise predictions for complex urban use classification. Consequently, in this article, we consider exploring a multitask supervision strategy to assist semantic feature description and regularize each semantic modality to locate its specific states in the proposed FSCNet.

III. Proposed Methodology

In this section, we first clarify the problem formulation and then describe the proposed FSCNet in detail. Then, Section III-A is the problem analysis, and Section III-B is an overview of the proposed FSCNet. Moreover, Section III-C describes the semantic component generation, Section III-D introduces the full semantic construction, and Section III-E elucidates the multitask supervision strategy.

A. Problem Analysis

In the single- and two-path methods of semantic feature description, encoding and decoding processes are important steps. Related to the semantic feature description, the studies of Liu et al. [24], Peng et al. [27], Li et al. [28], Ding et al. [30], Chen et al. [40], and Mou et al. [44] have demonstrated that hierarchical context information can be extracted via different scales of ERFs in the encoder and then used for semantic feature generalization in the decoder. In an ideal situation, ERF of each kind of scale can capture its corresponding context information, such as the large-scale ERFs that should capture the aggregated and abstract global information, the middle-scale ERFs that should capture the structure information, and the small-scale ERFs that should capture the boundary or

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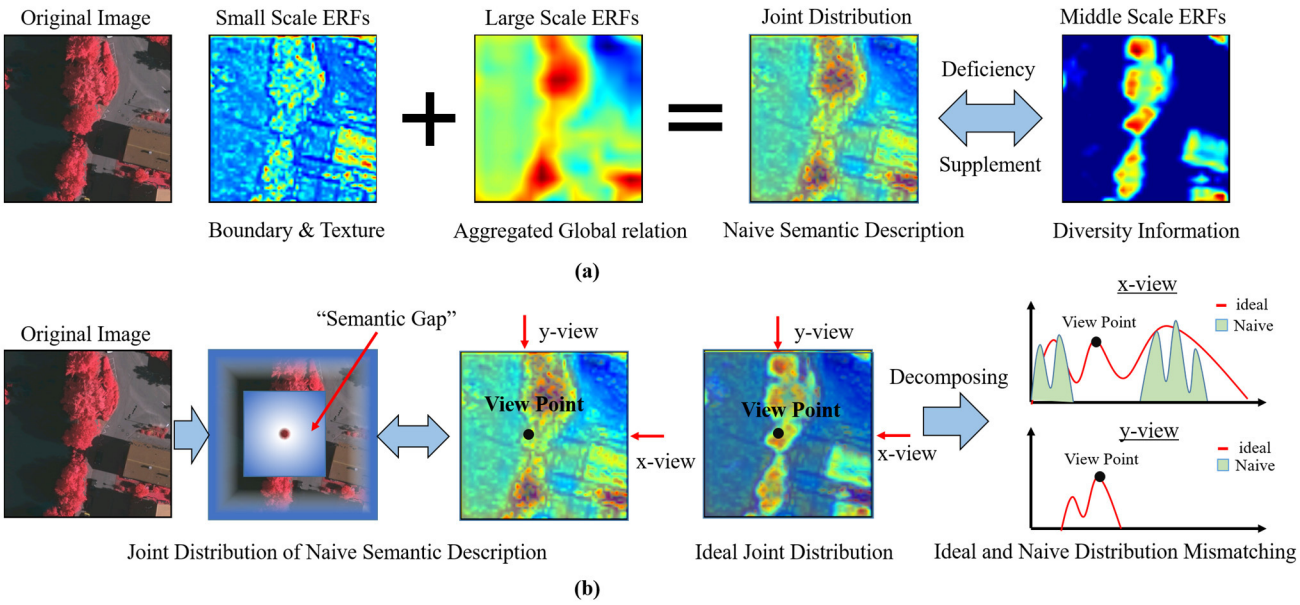


Fig. 2. Analysis of naive semantic description with a large number of small-scale ERFs and fewer ERFs that are not large enough. (a) Formation of naive semantic description. Deficiencies and Gaussian ERFs distributions mismatching between naive and ideal semantic descriptions.

texture information. When the valid boundary, structure, and abstract global information are reasonably combined, it can aggregate the full semantic information and form the perfect semantic feature description to support the arbitrary pixelwise prediction scenario.

However, we can find that, in Fig. 1, there is an unreasonable distribution of theoretical receptive fields (TRFs) along the CNN layers in the encoder, which cannot generate the diverse scales of ERFs to capture the effective hierarchical context information according to the following equation:

$$l_k = l_{k-1} + \left((f_k - 1) \cdot \prod_{i=1}^{k-1} s_i \right) \tag{1}$$

where l indicates the size of the squared TRFs in the convolutional layer and k is the index of the convolution layers. f_k indicates the convolution kernel size, which is usually set as an odd number. In addition, s_i is the convolution stride at the current layer. Then, following the rule of stacked convolution layers to enlarge TRFs, when s_i gradually becomes a large number and CNNs go to deeper layers, from Fig. 1(a) to (c), abundant small-scale and fewer large-scale TRFs are generated in the encoder. Then, according to (1), s_i directly impacts the changed trend of TRFs in the encoder for different CNN architectures (e.g., ResNet-18, ResNet-50, or ResNet-101). In Fig. 1(a)–(c), the green lines on the x-axis indicate the small-scale TRFs, which occupy a large proportion of convolution layers in the encoder, and the red and blue lines on the x-axis represent middle- and large-scale TRFs with a lower proportion of convolution layers in the encoder. Subsequently, Luo et al. [56] demonstrated that the ERFs are a Gaussian distribution and are located in squared TRFs to capture effective information. In addition, ERFs maintain the same change trend but are smaller than TRFs at each convolution layer. Thus, related to Fig. 1, there are only a large number of small-scale ERFs and fewer ERFs that are not large enough in the encoder, and they cannot extract valid hierarchical context information for better semantic description and pixelwise interpretation at the decoder.

As illustrated in Fig. 2(a) and according to the analysis of the ERF distribution in a normal CNN-based encoder, as shown in Fig. 1, the decoding process mainly focuses on combining the ERFs from the encoder to form the semantic description, which sets up a naive semantic description caused by only containing the ERFs that are small and not large enough. Subsequently, the naive semantic description is not suitable for more complex semantic relation descriptions, especially for pixelwise urban use classification. Here, let us assume that the ideal semantic description can be formulated as a joint distribution of 2-D Gaussian ERFs by referring to [56], and the ideal semantic description can be expressed as follows:

$$P(x) = \sum_{k=1}^{K} w_k \cdot \frac{1}{2\pi |\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(x - \mu_k)^T \Sigma^{-1} (x - \mu_k) \right) \tag{2}$$

where $p(x)$ is the joint distribution of 2-D Gaussian ERFs, and K is the index of diverse ERFs. w_k represents the weight of each 2-D Gaussian ERF, and x represents a sample of pixels, which can be captured by any scale of 2-D Gaussian ERFs. Σ is the covariance, and μ_k indicates the mean value of the kth 2-D Gaussian ERF component. Consequently, the naive semantic description can be roughly considered as having k equal to 1 or 2 (i.e., containing a large number of small-scale ERFs and fewer not large enough ERFs), which lacks the abundant scales of ERFs to capture the valid boundary, diverse structure, and abstract global information, as shown in Fig. 2(a). Then, the naive semantic description can also be

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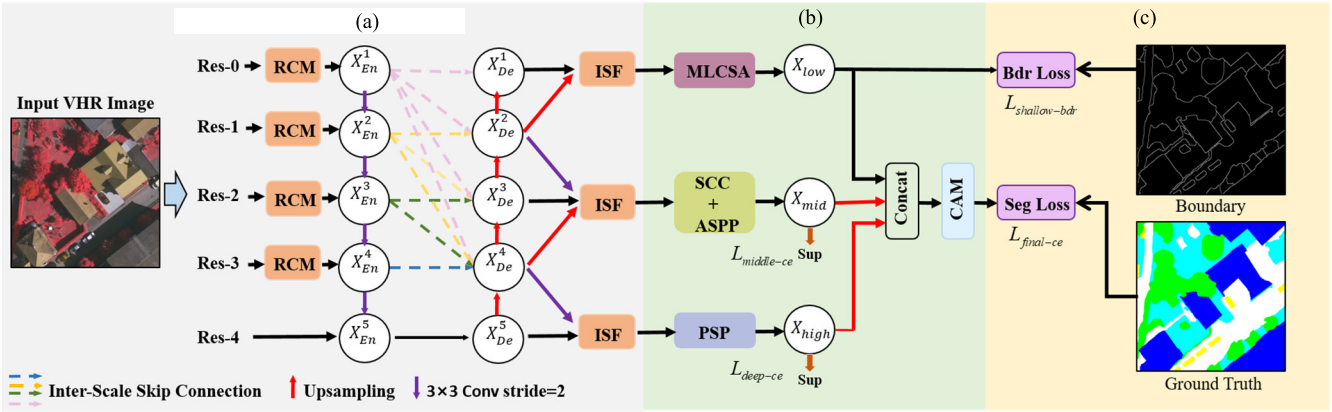


Fig. 3. Framework of the proposed FSCNet. (a) Semantic component generation. (b) Full semantic construction. (c) Multitask task supervision.

formulated as follows:

$$\begin{aligned} P'(x) &= P_1(x_1) + P_2(x_2) \\ &= w_1 \cdot \frac{1}{2\pi|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(x_1-\mu_1)^T \Sigma^{-1}(x_1-\mu_1)\right) \\ &\quad + w_2 \cdot \frac{1}{2\pi|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(x_2-\mu_2)^T \Sigma^{-1}(x_2-\mu_2)\right). \end{aligned} \tag{3}$$

Based on (2) and (3), if $K \gg 2$, there are considerable deficiencies between naive and ideal semantic descriptions, generating the severe semantic gap for pixelwise interpretation, as illustrated in Fig. 2(b). On the right-hand side of Fig. 2(b), the naive semantic description misses some view points by the naive 2-D joint distribution in (3); therefore, it usually provides large classification and localization errors for pixelwise prediction, which can be represented as the covariance Σ and mean μ_k. Here, the covariance Σ and mean μ_k can be decomposed into two 1-D Gaussian distributions with variance δ and mean μ_k from the horizontal x-view and the vertical y-view. Next, as shown in Fig. 2(b), from both the x-view and the y-view, Kullback-Leible divergence (KLD) [57] is utilized to evaluate the distribution similarity between the ideal and naive semantic feature descriptions by $D(P_{x-\text{view}}(x)||P'_{x-\text{view}}(x))$ and $D(P_{y-\text{view}}(x)||P'_{y-\text{view}}(x))$. Then, related to the two views of 1-D Gaussian distribution similarity, w_k weights combined with variance δ and mean μ_k should be major parameters used to measure the ability of classification and localization. In summary, owing to the unreasonable distribution of ERFs leading to naive semantic descriptions containing fewer w_k and unique variance δ and mean μ_k (i.e., a large number of small-scale ERFs and fewer ERFs that are not large enough), the valid or invalid boundary and interior texture information are randomly captured by small-scale ERFs. In addition, since fewer ERFs are not large enough in naive semantic descriptions, it cannot extract the effective abstract global information to set up a better semantic description from complex urban scenes. Therefore, to set up a more effective semantic description for refined urban use classification, we focus on studying the decoding process and introducing extra modules to rectify the joint distribution of 2-D Gaussian ERFs at the decoder. Then, considering randomly captured valid or invalid boundary and interior texture information according to small-scale ERFs, we set up a shallow semantic modality to specifically guide small ERFs capturing valid boundary information. Then, facing fewer ERFs that are not large enough, we rebuild the deep semantic modality to aggregate the abstract global information for final semantic feature descriptions. Besides, in view of the severe semantic gap between redefined shallow and deep semantic modalities, the middle semantic modality is also redefined to gracefully bridge the semantic gap and ensure the full semantic description for precise pixelwise prediction from VHR urban scenes.

B. Overview of FSCNet

In Fig. 3(a), the encoder–decoder of the proposed FSCNet is similar to that of Unet3+ [21] because it adopts abundant cross-scale connections, and NAS-FPN [58] demonstrated that abundant cross-scale connections can improve the semantic feature description and facilitate the superposition of full-scale ERFs for urban use classification. Yet, within the U-shaped encoder–decoder, there are many shallow layer features from the encoder to be introduced into the deep layer features of the decoder. As mentioned in [21] and [49] and analysis in Section III-A, enormous noise is introduced into the decoder, which worsens the semantic description for urban use classification. Therefore, related to the encoder–decoder of Unet3+ [21], we consider optimizing the shallow layer features from the encoder by designing an RRC module, which can mitigate noise from shallow layer features before abundant cross-scale connections. In addition, after abundant cross-scale connections, to advance different scale feature fusion by ERF superposition at the decoder, as shown in Fig. 3(a), a multiscale fusion (MSF) module is designed to achieve the different scale feature fusion and generate fused features of X^2_{De}, X^3_{De}, and X^4_{De}. After optimizing the encoder–decoder, the large difference between Unet3+ [21] and the proposed FSCNet is the final semantic feature fusion method. Unet3+ [21] utilizes dense connections to fuse several semantic features with different scales without considering their interscale interferences when directly fusing the different scale semantic features with a large-scale span. Here, interscale interferences among the different scale semantic features are

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considered in the proposed FSCNet, and an ISF module is designed to fuse several adjacent scale semantic features for initializing three semantic modalities because adjacent scale features can be considered to have the more relevant context information. Then, as illustrated in Fig. 3, after ISF, the shallow, middle, and deep semantic modalities are initialized to redefine a more effective semantic feature description. Considering the unreasonable ERF changes in the general CNN architecture, the invalid boundary or interior texture information is randomly captured by a large number of small-scale ERFs and fewer ERFs that are not sufficiently large to extract the insufficient abstract global information, which generates naive semantic descriptions. Therefore, MLCSA, SCC + ASPP, and PSP [19] modules are individually designed for initialized shallow, middle, and deep semantic modalities to rectify their ERFs and enhance the ability to acquire different semantic modality information. As shown in Fig. 3(b) and (c), for full semantic construction, the designed MLCSA and boundary supervision are applied for the initialized shallow semantic modality to assist in describing the pure and valid boundary. Then, PSP [19] and deep auxiliary supervision are used for the initialized deep semantic modality to further enlarge ERFs and capture more abstract global information. In addition, according to rebuilt shallow and deep semantic modalities, there are severe semantic gaps between pure boundary and more abstract global information. Therefore, the SCC + ASPP and deep auxiliary supervision are designed for the initialized middle semantic modality, which can allow the ERFs to make the middle semantic modality have quite good classification and localization ability, meanwhile, gracefully bridge the semantic gap between shallow and deep semantic modalities, and then facilitate the final semantic feature fusion. Finally, through full semantic construction, a convolutional block attention module (CAM) [59] is employed to adaptively select different semantic modalities (e.g., shallow, middle, and deep semantic modalities) for coupled pixelwise semantic relation descriptions. Next, in the training phase, boundary supervision and normal ground truth (GT) are used for multitask learning to help each semantic modality be located in its specific state. As follows, the three parts, including semantic component generation, full semantic construction, and multitask supervision strategy, are elucidated.

C. Semantic Component Generation

In Fig. 3(a), the cross-scale skip connections can incorporate smaller-scale ERFs from the encoder and larger-scale ERFs from the decoder to enrich ERF superposition with different scales. In each scale layer of the decoder, the cross-scale skip connections can be expressed as (4), shown at the bottom of the page, where X_{En}^{k} and X_{De}^{k} represent the kth stages in the encoder and the decoder; $D(\cdot)$ and $U(\cdot)$ are the downsampling and upsampling operations, respectively; $\ell(\cdot)$ and $[\cdot]$ are the convolution and concatenation operations, respectively; and $H(\cdot)$ realizes feature map aggregation with different scales of ERFs. Thus, at different decoder scales, ERF superposition not only introduces the corresponding scale encoding features into the decoder but also introduces previous scale encoder features with smaller ERFs. As shown in Fig. 3(a), except for the last scale of the decoder, the dashed lines with different colors represent different scales of ERFs, which correspond to the encoder features of X_{en}^{1}, X_{en}^{2}, X_{en}^{3}, and X_{en}^{4}. Then, a different ERF superposition can generate the semantic feature containing different semantic information, and according to different ERF combinations in the decoder, they can be used to build each semantic modality and set up full semantic descriptions.

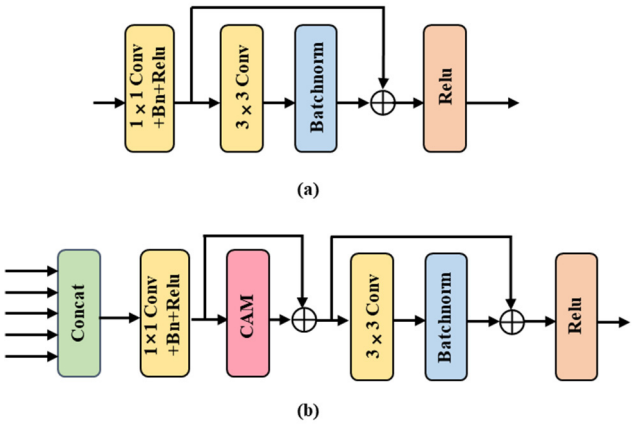


Fig. 4. Details of (a) designed RRC module and (b) designed MSF module.

1) Optimized Unet3+ Framework: Before encoding features are introduced into the decoder, an RRC module is designed to optimize encoding features from the backbone, as shown in Fig. 4(a). Since there are no shallow features introduced into X_{en}^{5}, the designed RRC is applied for other encoding features of X_{en}^{1}, X_{en}^{2}, X_{en}^{3}, and X_{en}^{4} from backbones Res-0, Res-1, Res-2, and Res-3. They will be individually fed into a 1×1 convolution with batch norm (BN) and the activation function of rectified linear unit (ReLU); then, they pass through a 3×3 convolution with BN and add themselves back before ReLU output. By applying the designed RRC module, a residual structure of a 3×3 convolution can make each feature layer further aggregate the spatial information and squeeze the channels to relieve random noise or interference to some extent. In addition, during the cross-scale skip connections in Fig. 3(a), by the cross-scale skip connections, the hierarchy encoding features are introduced into the decoder, and next, an MSF module is designed to compound ERFs into the decoding features to generate X_{De}^{2}, X_{De}^{3}, and X_{De}^{4}, as shown in Fig. 4(b). Related to the designed MSF module, each feature of the decoder and introduced encoding features

$$X_{\mathrm{De}}^{i} = \begin{cases} H\left(\left[\ell\left(D\left(X_{\mathrm{En}}^{k}\right)\right)_{k=1}^{i-1}, \quad \ell\left(X_{\mathrm{En}}^{i}\right), \quad \ell\left(U\left(X_{\mathrm{De}}^{i+1}\right)\right)\right]\right), & i = 1, \ldots, N-1 \\ X_{\mathrm{En}}^{i}, \quad i = N \end{cases} \tag{4}$$

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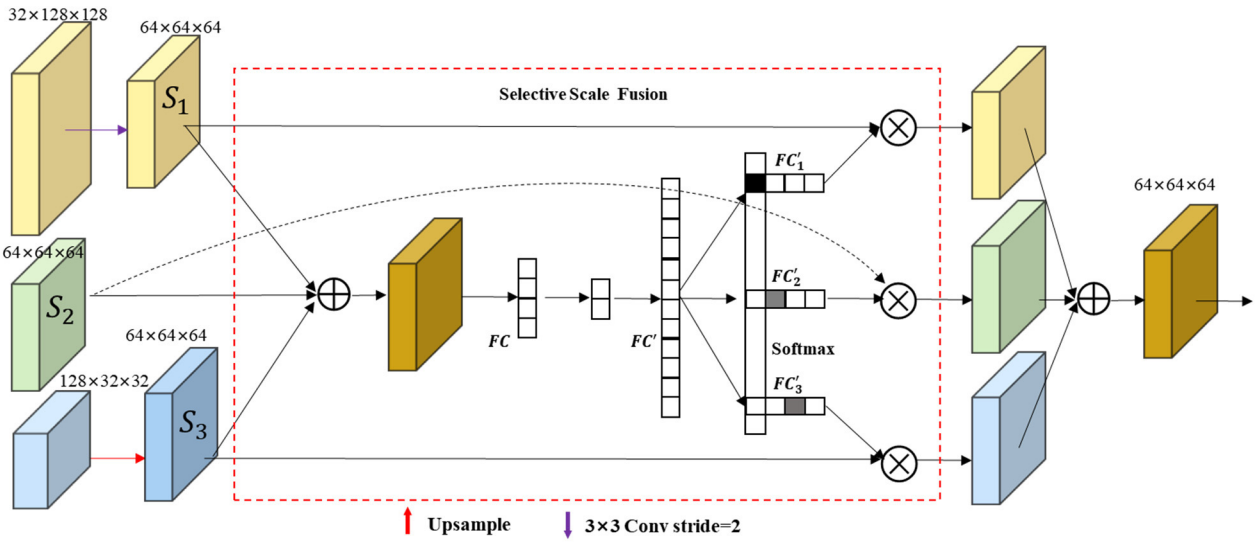


Fig. 5. Detail of the designed ISF module.

are concatenated and fused by a 1 × 1 convolution, BN, and ReLU, and then, CAM [59] is utilized to adaptively select the more useful features. Next, a 3 × 3 convolution with BN and ReLU is used for channel selected features to further restrain the noise or interference. Finally, after the optimized Unet 3+ [21] encoder–decoder, the ERFs are reasonably combined.

2) ISF Module: To gradually fuse adjacent scale semantic features, the ISF module is proposed to generate the initial semantic modalities, as shown in Fig. 5. The inputs of the designed ISF are three or two different scale features from the decoder. Then, X^1_{De} and X^2_{De} are both considered the shallow semantic modality. X^2_{De}, X^3_{De}, and X^4_{De} can be seen as the middle semantic modality, whereas X^4_{De} and X^5_{De} can be regarded as the deep semantic modality. Next, to add adjacent semantic features, they are both regulated into a uniform size by using the downsampling and interpolation operations, as shown in Fig. 5. Here, downsampling refers to a 3 × 3 convolution with a stride = 2. When adjacent scale semantic features are regulated into the same size, they will be fed into the designed ISF module to adaptively facilitate multiscale feature fusion. As an example shown in Fig. 5, after three adjacent semantic features are added together, global average pooling is employed to generate a global feature vector FC. Subsequently, two 1 × 1 convolutions are employed to generate the channelwise feature weight of FC′. Next, a softmax function is applied for FC′ by channel splitting, which can produce the corresponding weights of FC'_1, FC'_2, andFC'_3 to selectively reweight the different scale semantic features for the final fusion, and it can be expressed as follows:

$$\begin{aligned} F_{\mathrm{fusion}} &= \alpha_1 S_1 + \alpha_2 S_2 + \alpha_3 S_3 \\ &= \frac{e^{\mathrm{FC}'_1}}{e^{\mathrm{FC}'_1} + e^{\mathrm{FC}'_2} + e^{\mathrm{FC}'_3}} \cdot S_1 + \frac{e^{\mathrm{FC}'_2}}{e^{\mathrm{FC}'_1} + e^{\mathrm{FC}'_2} + e^{\mathrm{FC}'_3}} \cdot S_2 \\ &\quad + \frac{e^{\mathrm{FC}'_3}}{e^{\mathrm{FC}'_1} + e^{\mathrm{FC}'_2} + e^{\mathrm{FC}'_3}} \cdot S_3 \end{aligned} \tag{5}$$

where F_{fusion} represents the final fused features; α_1, α_2, and α_3 are adaptively learned coefficients of FC'_1, FC'_2, and FC'_3, which can achieve selective semantic feature combination; and S_1, S_2, and S_3 are inputs of regulated semantic features with different scales. Then, following (5), adjacent scale semantic features from the encoder–decoder can be fused by their corresponding learned coefficients. Finally, by the designed ISF module, the initial shallow, middle, and deep semantic modalities can be formulated and prepared for full semantic construction.

D. Full Semantic Construction

As mentioned above, considering the unreasonable ERF distribution leading to naive semantic description for a limited pixelwise interpretation ability, a more effective semantic method of full semantic construction is proposed to rebuild the three specific modalities (e.g., shallow, middle, and deep semantic modalities), which can improve the naive semantic description and then ensure the completeness and effectiveness for more complex urban scene interpretation.

1) Shallow Semantic Modality: Owing to the naive semantic description, there are several small-scale ERFs that would randomly capture the valid or invalid boundary or interior texture to degrade shallow semantic features. Although some noise is considered to be suppressed in the encoder–decoder architecture of the proposed FSCNet, owing to the limited receptive fields of the initialized shallow semantic modality, it is also difficult to ensure that the shallow semantic modality captures basic and valid boundary information. Consequently, an MLCSA module combined with boundary supervision is designed to guide shallow semantic modality to correctly capture the basic, pure, and valid boundary information from complex urban scenes. As shown in Fig. 6, the high-spatial-resolution feature of the initial shallow semantic modality is fed into the designed multiscale context feature extraction structure. Here, to enlarge ERFs and obtain abundant context

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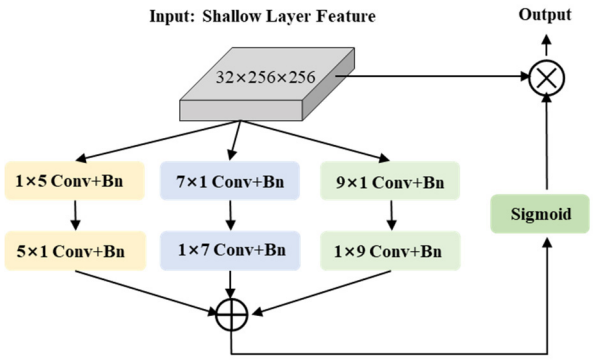


Fig. 6. Detail of the designed MLCSA module.

information, the 5 × 5, 7 × 7, and 9 × 9 convolutions are applied for each channel of input to capture the spatial context concerns. Then, the extracted multiscale context information is compounded and normalized by addition and sigmoid function. Currently, context-guided information is produced by active multiscale context information with GT supervision, and it can also be used to rectify the initial shallow semantic modality by context reweighting. Then, much invalid boundary or interior context information can be eliminated. Finally, by using the designed MLCSA module and boundary supervision, the basic, pure, and valid boundary can be captured by the shallow semantic modality and beneficial for localization ability improvement for full semantic construction. In addition, it can also provide accurate boundary prediction from complex urban scenes.

2) Deep Semantic Modality: Related to the naive semantic description, there are also a few ERFs that are not large enough for pixelwise interpretation, which makes several incorrect classification situations affect urban use interpretation ability. Thus, the PSP [19] module is employed to further enhance the semantic feature generalization ability for the initial deep semantic modality. Four different sizes of pooling operations are applied for the initial deep semantic modality to further enlarge the receptive field and capture more abstract global information, also called long-range dependence. Then, after the PSP [19] module is applied for the deep semantic modality, it can improve the discriminant ability of global context information with a large receptive field and leverage the final full semantic construction.

3) Middle Semantic Modality: Based on the rebuilt shallow and deep semantic modalities, there is a severe semantic gap between the shallow and deep semantic modalities, which leads to the fused semantic feature not balancing the performance of classification and localization. Thus, to bridge the severe semantic gap, a middle semantic modality is considered, which is the most important process for full semantic construction. The SCC combined with the ASPP [15] module is designed to rectify the initial middle semantic modality to make it have better classification and localization ability to replenish the shallow and deep semantic modalities. As shown in Fig. 7, the initial middle semantic modality M is individually fed into two 1 × 1 convolutions with BN and ReLU to generate two modalities $\{M_1, M_2\}$. Subsequently, each of them would be simultaneously fed into two branches of the designed SCC as input. In the top branch of Fig. 7, to improve localization ability, the input M_1 would downsample r times, and then, after a 3 × 3 convolution, BN, and ReLU, it would upsample r times to obtain complete context information. In Fig. 7, the upsampling feature of M_1^1 is added back into the input M_1 to generate M_1^2, and a sigmoid function is applied for M_1^2 to produce a spatial context coefficient map of M_1^4. Meanwhile, 3 × 3 convolution, BN, and ReLU operations are applied for M_1 to generate the equivalent feature M_1^3; then, the spatial context coefficient M_1^4 can be used to reweight the spatial regions and improve its pixelwise localization ability. These above operations can be formulated as follows:

$$M_1^2 = M_1^1 + M_1 = \mathrm{Up}\big(T_1\big(\mathrm{AvgPool}_r(M_1)\big)\big) + M_1 \quad (6)$$

$$M_1^3 = T_2(M_1) \quad (7)$$

$$M_1^4 = M_1^3 \odot \sigma\big(M_1^2\big) \quad (8)$$

where $\mathrm{Up}(\cdot)$ is a bilinear interpolation operation, σ is the sigmoid function, $T(\cdot)$ denotes a 3 × 3 convolution followed by BN and ReLU operations, and $+$ and $\odot$ denote element-wise addition and multiplication, respectively. Then, following (6)–(8), as shown in Fig. 7, the localization ability of a part of M_1 can be improved. In another branch, for the given input M_2, a CAM [59] is performed, yielding a channel reweighted feature M_2^1. Finally, localization and classification calibrated features of M_1^4 and M_2^1 are concatenated and fused by the 3 × 3 convolution, BN, and ReLU operations, yielding the output M_{out}, M_2^1, and M_{out}, which can be, respectively, expressed as follows:

$$M_2^1 = \mathrm{CA}(M_2) \quad (9)$$

$$M_{\mathrm{out}} = T\big(\big[M_1^4, M_2^1\big]\big) \quad (10)$$

where $\mathrm{CA}(\cdot)$ represents the CAM [59]. Following (9) and (10), when the initial middle semantic modality is rectified by the designed SCC module, it can obtain both good location and classification abilities for pixelwise interpretation. Moreover, after the SCC module, the compound ERFs of an initial middle semantic modality are rectified. Then, as shown in Fig. 2(b), to better supplement the full semantic description, the ASPP [15] module is employed to expand the multiscale representation of the generated middle semantic modality M_{out}, which can gracefully bridge the severe semantic gap between shallow and deep semantic modalities for full semantic construction.

In the end, after middle, shallow, and deep semantic modality optimization, the more completed semantic description can be rebuilt. The shallow semantic modality is redefined to capture valid boundary information to improve pixelwise localization ability. Then, the deep semantic modality mainly improves the generalization ability of the semantic feature description to capture more abstract global information, and then, the middle semantic modality is considered a supplement to bridge the semantic gap between shallow and deep semantic modalities. Then, related to these three semantic modalities, the middle and deep semantic modalities are upsampled into the same size as the shallow semantic modality, and after the concatenation of shallow, middle, and deep semantic

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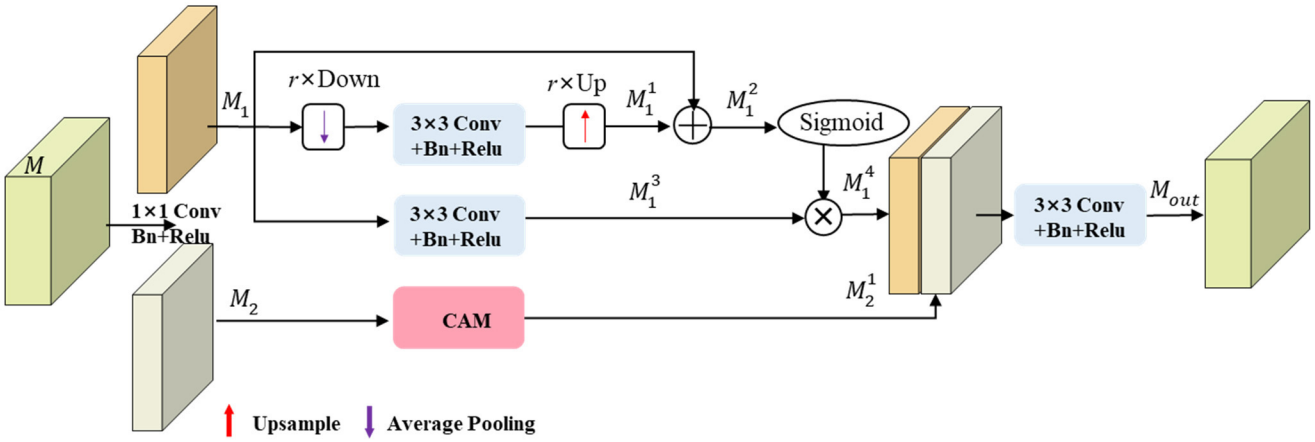


Fig. 7. Detail of the designed SCC module.

modalities, CAM [59] is used to achieve adaptive semantic modality selection, which can facilitate fine-grained pixelwise urban use classification.

E. Multitask Supervision Strategy

To leverage each semantic modality to obtain valid information in the implicit CNN feature extraction paradigm, a multitask supervision strategy is designed to make each redefined semantic modality locate in its specific state and facilitate the final refined urban use classification. As shown in Fig. 3(c), the boundary information is used to supervise the shallow semantic modality and enable it to capture the valid boundary information while avoiding noise or interference during the training phase. The boundary loss is described as follows:

$$L_{\text{shallow-bdr}} = \sum_i \text{BCE}\left(b^{\wedge}, b\right) = -\frac{1}{m}\sum_i^m \times \left[\lambda_1 \cdot b_i \log\left(b_i^{\wedge}\right) + \lambda_2(1-b_i)\log\left(1-b_i^{\wedge}\right)\right] \tag{11}$$

$$\lambda_1 = \frac{B^0}{B^1 + B^0}\lambda_0 = \frac{B^1}{B^1 + B^0} \tag{12}$$

where m denotes the sample number and $\hat{b}$ is the predicted boundary; b denotes the boundary GT that comes from the semantic label; and B^0 and B^1 denote the numbers of nonboundary and boundary pixels, respectively. Moreover, we apply the standard cross-entropy loss to supervise middle semantic modality, deep semantic modality, and final semantic prediction of urban use classification. The cross-entropy loss is calculated as follows:

$$L_{\text{ce}} = -\frac{1}{m}\sum_i^m \sum_j^C \log P\left(y^i = j | x^i\right) \tag{13}$$

where m and C denote the number of samples and categories; $x^i \in \mathbb{R}^m$ denotes the ith sample; $y^i = j (j \in \{1, \ldots, C\})$ denotes the pixels belonging to the jth class label; and P represents the probability score of classification prediction. Let the middle semantic supervision loss be denoted $L_{\text{middle-ce}}$, the deep supervision loss be denoted $L_{\text{Deep-ce}}$, and the final semantic supervision be denoted $L_{\text{final-ce}}$; furthermore, the multitask supervision loss can be expressed as follows:

$$L_{\text{Multi-Task}} = L_{\text{final-ce}} + L_{\text{shallow-bdr}} + L_{\text{middle-ce}} + L_{\text{deep-ce}}. \tag{14}$$

Following (15), the multitask supervision losses can promote each semantic modality to effectively facilitate the redefined full semantic construction.

IV. Experiments and Analysis

To validate the effectiveness of the proposed FSCNet, we performed extensive experiments on three prominent aerial image semantic segmentation datasets, namely, the ISPRS 2-D semantic labeling benchmark Vaihingen and Potsdam datasets [51] and the DLRSD [52] dataset. In this section, we first introduce the datasets, experimental settings, and evaluation metrics. Afterward, several relative SOTA models were chosen for comparison and tested on the three datasets. In addition, we perform the ablation study to analyze the contribution of each component in the proposed model.

A. Datasets

To evaluate the performances of the proposed FSCNet and SOTA models, three urban use classification datasets, ISPRS Vaihingen [51], ISPRS Potsdam [51], and DLRSD [52], were employed for training, validation, and testing. To present the experimental conditions in detail, three datasets are described as follows.

1) Potsdam: This dataset shows a typical historic city with large building blocks, narrow streets, and dense settlement structures. It comprises 38 tiles (6000 × 6000 pixels), and each tile comprises four channels, near-infrared (NIR), red (R), green (G), and blue (B), with a 5-cm spatial resolution. In our experiments, 17 tiles were selected as the training set, seven tiles (areas: 2_11, 2_12, 4_10, 5_11, 6_7, 7_8, and 7_10) were selected as the validation set, and the remaining 14 tiles were used to evaluate the generalization ability of the models.

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2) Vaihingen: This dataset shows a relatively small village with many detached buildings and small multistory buildings. It comprises 33 tiles with varying sizes (e.g., the average size is 2494 × 2064 pixels), and each tile comprises three channels NIR, R, and G, at a 9-cm spatial resolution. In the experiments, 16 of 33 tiles were selected for training: 11 tiles for training, five tiles (areas: 11, 15, 28, 30, and 34) for validation, and the other 17 tiles for testing.

Potsdam and Vaihingen are two urban scene datasets, and they have been labeled with six of the most common urban use classes: impervious surface (IS), buildings (B), low vegetation (LV), trees (T), cars (C), and clutter (CL). These two datasets include the VHR true orthophoto (TOP) tiles and the corresponding digital surface models (DSMs) derived from dense image matching techniques; taking into account the inaccessibility and nonuniversality of DSM data, the DSMs were not used in the experiments.

3) DLRSD: This is a dense labeling dataset and is an extension of the UCMERCED dataset [60], which was constructed for multilabel image retrieval. DLRSD can be used for multilabel tasks, such as scene-based tasks (e.g., remote sensing image retrieval and classification) and other pixel-based tasks (e.g., semantic segmentation). The images and class numbers in DLRSD [52] are the same as those in UCMERCED [60]. There are 21 broad categories with 100 images per class; these images were manually extracted from large images from the USGS national map urban area imagery collection for various areas around the country. The pixel resolution of this public domain imagery is 1 ft, and each image is labeled pixelwise based on 17 predefined class labels, such as the airplane, baresoil, buildings, cars, chaparral, court, docks, field, mobilehome, grass, pavement, sand, sea, ship, tanks, trees, and water.

B. Data Augmentation

For the ISPRS Vaihingen and Potsdam datasets [51], we employ the sliding window method to cut the original images from the top left to the bottom right to small patches with sizes of 512 × 512 pixels with a 50% overlap rate. To improve the generalization ability of models, the diversity and quantity of training samples should be extended by data augmentation. We applied random horizontal flipping, random scaling from 0.5 to 2.0, and random cropping to 512 × 512 pixels for all training images. During testing, sliding window sizes of 512 × 512 pixels at a 128-pixel stride were used on the testing images. The results were stitched together by averaging the predictions of the overlapping region to create the final prediction output. Then, regarding the DLRSD dataset [52], data argumentation methods, such as random horizontal flipping, random scaling from 0.5 to 2.0, and random cropping into 256 × 256 pixels, were used for all training images. Because the sizes of a few images were not exactly 256 × 256 pixels, we resized them to 256 × 256 first. In the experiments, 2100 DLRSD images were split into 1470 and 630 images for training and validation.

C. Evaluation Metric

To measure the quantitative performances of pixelwise urban use classification, we considered four widely used metrics, the overall accuracy (OA), pixel accuracy (PA), F1-score (F_1), and intersection over union (IoU) as evaluation indices, which are defined as follows:

$$\mathrm{OA} = \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{FP}+\mathrm{TN}+\mathrm{FN}} \tag{15}$$

$$\mathrm{PA} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \tag{16}$$

$$F_1 = \frac{2\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}+\mathrm{FN}} \tag{17}$$

$$\mathrm{IoU} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}+\mathrm{FN}} \tag{18}$$

where TP, FP, FN, and TN are the numbers of true positive, false positive, false negative, and true negative pixels from the prediction maps, respectively. Notably, mF1 and mIoU were computed as the average measure of all classes except the clutter class. In addition, the semantic GT was selected as the eroded version whose object boundaries were eroded by a 3-pixel radius. In addition, the Kappa index [61] was also employed to evaluate the accuracy of boundary prediction.

D. Experiment Setting

All experiments were implemented on the PyTorch 1.10.4 deep learning framework and conducted on a server computer equipped with two Intel Xeon Gold 6240 CPUs, two 32-GB RAM units, and four NVIDIA TITAN RTX GPUs. During training, the adaptive moment estimation (Adam) optimizer was employed. Then, for the learning rate schedule, we adopted a warm-up learning policy in the first 20 epochs, and the models were trained for 80 epochs with an initial learning rate of $1e-4$ and the learning rate policy scheduled by a decaying factor of 0.9. The batch size of each model was set in the dynamic range of 6–4.

E. Comparative Analysis of Results

To comprehensively compare and analyze the refined urban use classification performance, first, several prominent and widely used semantic segmentation networks, including the FCN [14], DeepLabv3+ [15], Unet [17], GCN [18], PSPNet [19], Unet++ [20], and Unet3+ [21], were employed as comparisons, which almost covered the existing semantic generalization strategies. In addition, several recently proposed optical remote sensing urban use classification networks, including CANet [35], MACU [36], DDCM [24], BAMTL [48], and SBANet [49], and our previously proposed networks, FE-Net [29] and DI-Net [39], were used as comparisons to demonstrate the effectiveness of the proposed FSCNet. Next, all networks were tested on the three datasets, and the comparison results are described.

1) ISPRS Potsdam and Vaihingen: To fairly compare the selected networks, ResNet-101 [62] was utilized as a backbone for all comparisons. As shown in Tables I and II, the results indicated that the proposed FSCNet outperformed other SOTA methods, with a 90.53% OA, an 81.83% mIoU, and an 89.88% mF1 on ISPRS Vaihingen and a 91.16% OA, a 92.65% mIoU, and an 86.56% mF1 on ISPRS Potsdam. Then, in lines 1 and 2 of Tables I and II, because the FCN [14] and PSPNet [19]

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TABLE I
COMPARISONS OF SEGMENTATION PERFORMANCE ON ISPRS VAIHINGEN WITH NO BOUNDARY STATISTIC

Method	*IS*	*B*	*LV*	*T*	*C*	***OA***	***mF1***	***mIoU***
FCN [14]	92.75	**95.14**	81.29	91.97	70.50	90.13	87.23	77.98
PSPNet [19]	92.70	94.51	82.10	91.09	72.02	89.99	87.30	78.02
GCN [18]	93.09	94.46	79.85	92.71	85.68	90.04	89.18	80.73
Deeplabv3+[15]	92.20	94.69	81.25	91.71	87.65	89.99	89.28	80.88
CANet [35]	92.53	94.88	80.09	92.07	88.02	90.01	89.39	81.06
DDCM [24]	93.04	94.70	81.07	92.00	85.53	90.21	89.43	81.12
DI-Net [39]	91.94	94.70	81.90	92.09	87.29	90.08	89.23	80.77
FE-Net [29]	92.88	94.55	80.37	92.94	86.21	90.28	89.67	81.50
Unet [17]	92.96	93.96	80.17	**92.72**	88.00	90.00	89.15	80.68
Unet++ [20]	92.65	94.97	80.91	92.00	86.27	90.15	89.30	80.91
Unet3+ [21]	91.86	95.05	80.93	92.17	**88.98**	90.08	89.45	81.16
MACU [36]	93.11	94.87	79.97	92.74	87.91	90.28	89.55	81.31
SBANet [49]	91.94	95.11	**82.88**	90.85	86.71	90.20	89.51	81.25
BAMTL [48]	91.82	95.04	81.13	92.23	87.64	90.13	89.35	81.00
FSCNet (our)	**93.20**	94.50	82.33	91.76	88.06	**90.53**	**89.88**	**81.83**

TABLE II
COMPARISONS OF SEGMENTATION PERFORMANCE ON ISPRS POTSDAM WITH NO BOUNDARY STATISTIC

Method	*IS*	*B*	*LV*	*T*	*C*	***OA***	***mF1***	***mIoU***
FCN [14]	94.46	96.77	89.71	87.35	94.53	90.66	91.76	84.98
PSPNet [19]	94.54	97.24	89.15	**88.56**	92.52	90.97	91.89	85.16
GCN [18]	**95.34**	96.07	89.73	85.70	95.53	90.61	92.02	85.45
Deeplabv3+[15]	94.79	96.79	88.93	87.37	95.65	90.89	92.24	85.84
CANet [35]	94.10	95.77	**91.10**	85.45	96.26	90.59	92.10	85.59
DDCM [24]	94.36	96.84	88.38	88.14	95.80	90.67	92.25	85.85
DI-Net [39]	93.91	96.86	89.46	88.01	95.60	90.90	92.34	85.95
FE-Net [29]	94.36	96.61	89.98	87.76	96.37	90.93	92.48	86.23
Unet [17]	93.69	97.30	88.81	88.47	96.16	90.79	92.24	85.83
Unet++ [20]	93.92	96.90	89.97	86.41	96.27	90.80	92.30	85.94
Unet3+ [21]	94.45	97.30	88.78	87.10	95.86	90.87	92.36	86.05
MACU [36]	94.12	97.31	89.86	87.30	96.46	90.95	92.39	86.10
SBANet [49]	94.12	97.29	89.99	86.03	96.84	90.85	92.37	86.10
BAMTL [48]	93.53	**97.70**	88.86	88.02	96.73	90.81	92.30	85.97
FSCNet (our)	94.52	97.27	90.34	86.38	**96.98**	**91.16**	**92.65**	**86.56**

did not focus on fusing shallow layer features with small-scale ERFs, they produced poor results for the predictions of small cars compared with other methods. In addition, since they contain only relatively single-scale ERFs with naive semantic descriptions, they both performed poorly on the two ISPRS datasets. Next, since the cross-scale connections and ASPP were introduced into the encoder–decoder, the DeepLabv3+ [15], GCN [18], Unet [17], Unet++ [20], and Unet3+ [21] outperformed the FCN [14] and PSPNet [19]; Unet 3+ [21], in particular, in view of the superposition of its more abundant ERFs at each level of the decoder when modeling complex semantic relations, could obtain good performance with a 90.08% OA, an 81.16% mIoU, and an 89.45% mF1 on ISPRS Vaihingen and a 90.87% OA, an 86.05% mIoU, and a 92.36% mF1 on ISPRS Potsdam. Thus, the encoder–decoder strategy of Unet 3+ [21] was considered for plugging into the proposed FSCNet. However, in Unet3+ [21], there was enormous interference or random noise in shallow layer encoding features, which would be introduced into each scale of decoding features and severely affect the performance of dense connection feature fusion. Therefore, we optimized the encoder–decoder strategy of Unet 3+[21] to gradually generate different scales of semantic features more reasonably, and then, the full semantic description paradigm was designed to replace the dense connection feature fusion strategy, which made the proposed FSCNet outperform Unet3+ [21] on the two ISPRS datasets.

Furthermore, as reported in Tables I and II, several networks, including CANet [35], MACU [36], DDCM [24], BAMTL [48], SBANet [49], FE-Net [29], and DI-Net [39], which consider the requirement of refined urban use classification, were employed to further explore the more effective semantic description strategies. For example, BAMTL [48] and SBANet [49] individually tried to introduce boundary

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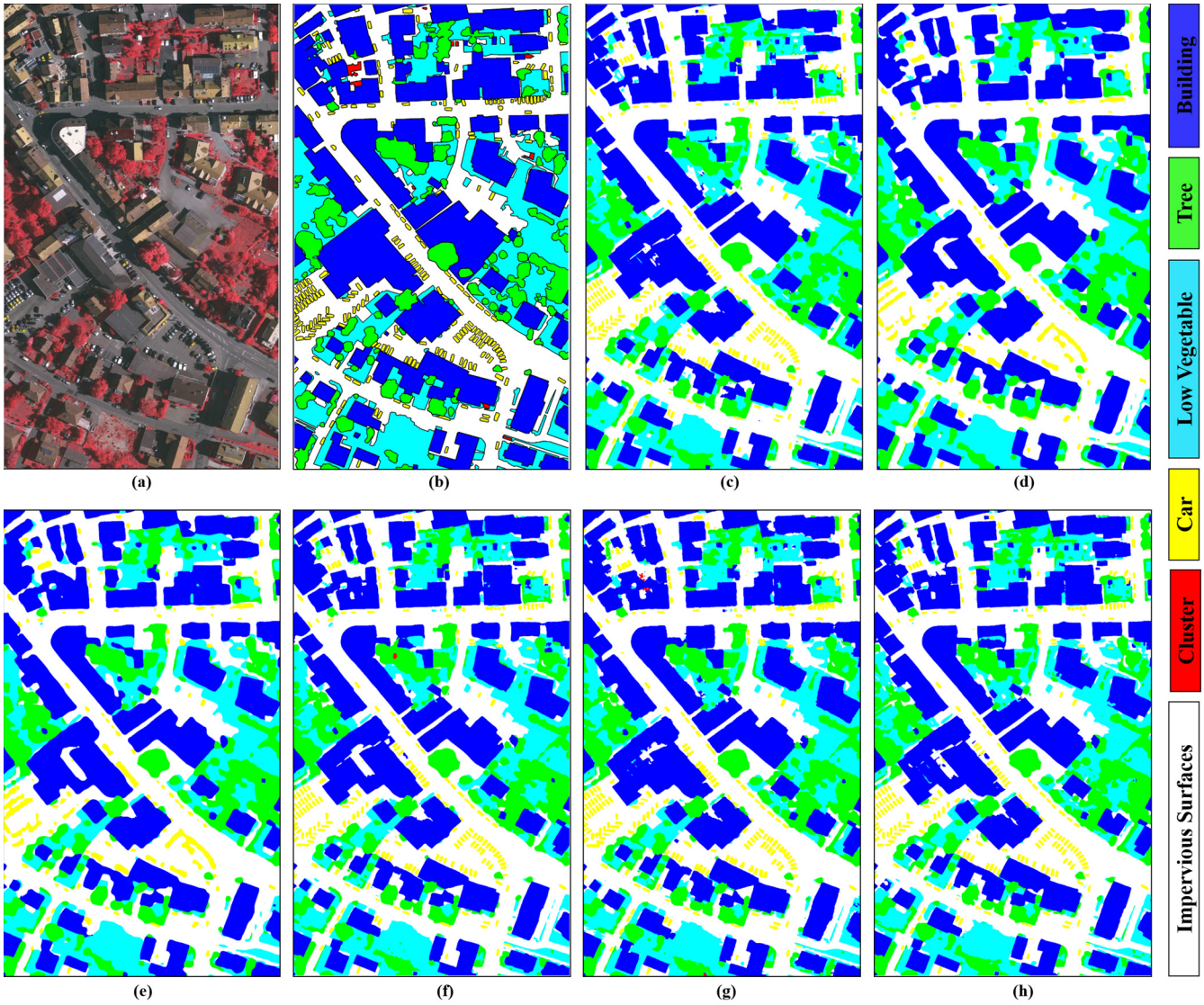


Fig. 8. Segmentation results on ISPRS Vaihingen. (a) Original image. (b) GT. (c) Result of the proposed FSCNet. (d) Result of FCN [14]. (e) Result of PSPNet [19]. (f) Result of MACU [36]. (g) Result of Unet3+ [21]. (h) Result of FE-Net [29].

information by defining a specific module and extra branch, which aimed to simultaneously promote localization and classification abilities. Then, CANet [35], MACU [36], and DDCM [24] employed the residual structure, parallel multiscale convolutions, different rates of dilated convolutions, and different types of attention structures in each network to enhance the complex semantic generalization ability. Although they could obtain competitive performance on the two ISPRS datasets, these networks could not handle random noises from shallow layer features and the naive semantic description of the low- and high-level feature fusions. Thus, these networks had lower OA, mIoU, and mF1 than the proposed FSCNet. In lines 7 and 8 of Tables I and II, our previously proposed networks, FE-Net [29] and DI-Net [39], also attempted to find optimized feature fusion and more reasonable encoder–decoder strategies to achieve refined urban use classification. However, DI-Net [39] only considered low- and high-level feature fusions by an improved attention mechanism, and there were considerable semantic differences or gaps between the low- and high-level features, which were difficult to rectify by the designed stage grading loss. Thus, DI-Net [39] only had good performance for fine-scale details but could not outperform the proposed FSCNet. Related to FE-Net [29], we found that the middle semantic modality deficiency would lead to poor results, which exhibited a natural contradiction of localization and classification predictions; thus, we introduced the attention mechanism to reweight a specific middle feature layer from the encoder and set up the middle semantic modality to make it have both good localization and classification abilities to some extent. However, the random noise or interference in the shallow feature layer severely affected the attention calculation and semantic feature fusion. In addition, because there was no abundant ERF combination, FE-Net [29] could not handle complex urban scenes more than the proposed FSCNet by using only the encoding features. In general, from Tables I and II, the richer

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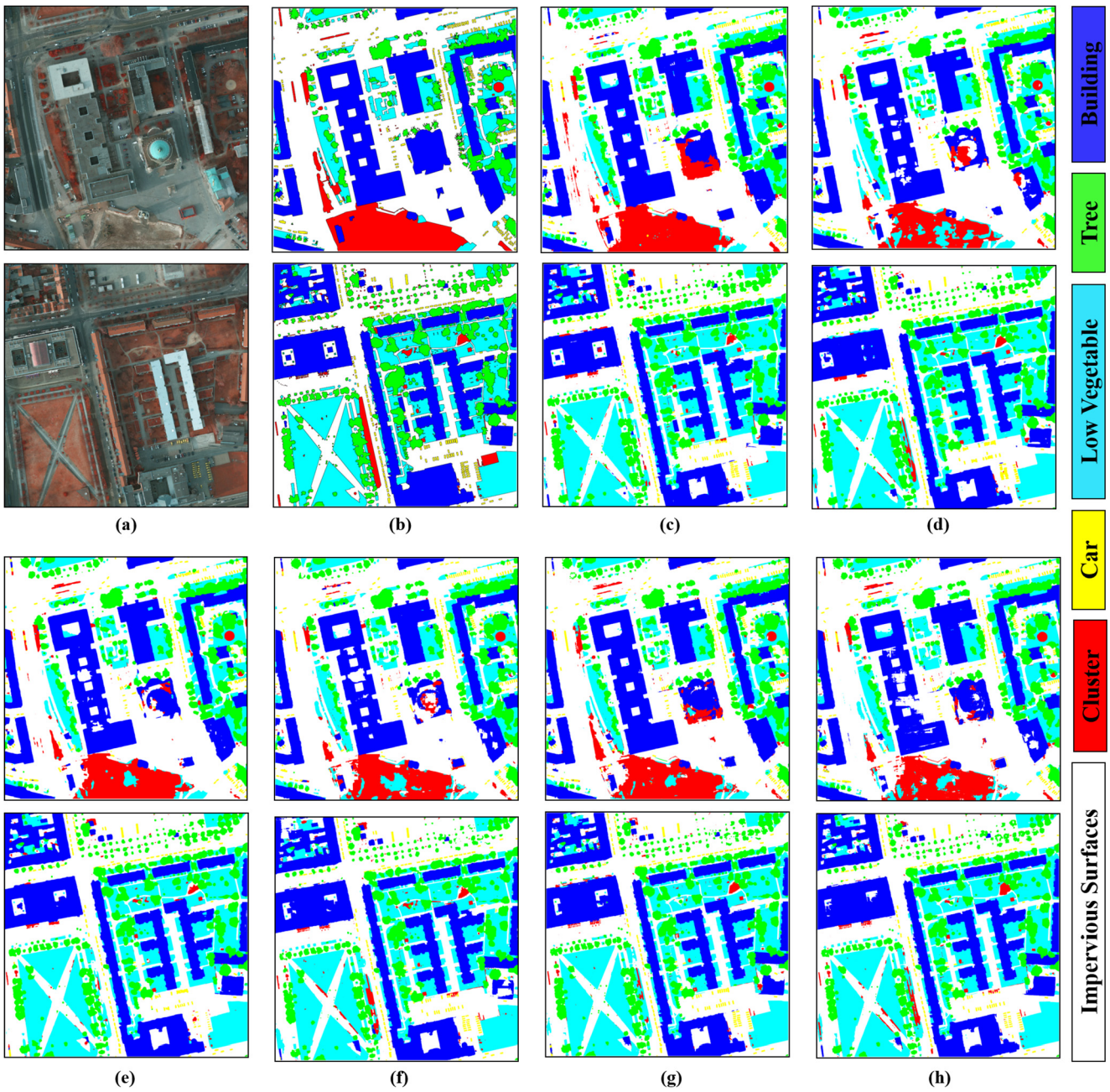


Fig. 9. Segmentation results (cropped images) on ISPRS Potsdam. (a) Original images. (b) GTs. (c) Results of the proposed FSCNet. (d) Results of FCN [14]. (e) Results of PSPNet [19]. (f) Results of MACU [36]. (g) Results of Unet3+ [21]. (h) Results of FE-Net [29].

and more sufficient ERF combinations, such as Unet 3+ [21] and MACU [36], performed fairly well. Furthermore, more reasonable semantic feature construction, such as FE-Net [29] and the proposed FSCNet, performed well, which reflects the superiority of abundant ERF superposition and the relevance of reasonable semantic feature construction. Next, several visualization results are shown in Figs. 8 and 9. In Figs. 8 and 9, the naive semantic description in the FCN [14] and PSPNet [19] could not provide good performance, while the more adequate semantic modality fusion and more reasonable full semantic description in the proposed FSCNet provided better results.

2) DLRSD: To further demonstrate that the proposed FSCNet could provide an effective full semantic description, the more challenging urban use classification dataset, DLRSD [52], which contained 17 pixelwise labeled classes of urban uses, was employed for comparisons. For a fair comparison, ResNet-101 [62] was applied to all compared networks, as reported in Table III, and 17 categories corresponded to numbers from 0 to 16. Regarding the comparison results, the proposed FSCNet outperformed other networks, with 81.64% OA, 67.31% mIoU, and 79.62% mF1. Owing to the more complex and variable semantic relations from the 17 categories of urban uses than those of ISPRS datasets, the generally and

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TABLE III
COMPARISONS OF SEGMENTATION PERFORMANCE ON DLRSD

Method	*mIoU*	*mF1*	*OA*	*0*	*1*	*2*	*3*	*4*	*5*	*6*	*7*	*8*	*9*	*10*	*11*	*12*	*13*	*14*	*15*	*16*
FCN [14]	51.01	64.56	72.84	12.56	48.52	74.94	46.05	61.23	90.96	38.32	92.78	67.01	32.51	86.74	56.37	92.19	48.02	69.84	78.27	85.98
PSPNet [19]	54.69	68.00	75.65	25.54	56.89	84.13	49.34	53.40	92.43	37.96	93.85	71.90	49.55	86.63	54.45	**96.34**	46.48	78.66	78.42	86.53
GCN [18]	62.85	75.72	79.76	35.63	63.52	86.22	81.59	78.95	89.17	79.50	93.99	72.73	38.80	88.88	52.62	90.44	65.73	**81.05**	**84.61**	89.84
Deeplabv3+[15]	63.92	77.05	80.19	44.17	65.06	84.25	83.01	81.25	91.01	75.65	84.13	74.87	48.97	89.50	61.72	95.12	70.52	72.94	81.50	88.52
CANet [35]	64.09	76.82	80.07	29.59	62.60	85.92	82.86	78.27	88.30	**81.25**	88.07	75.45	56.76	88.42	55.29	88.73	75.61	77.77	84.43	89.63
DDCM [24]	64.43	77.27	80.42	42.84	64.89	85.87	84.40	81.90	89.35	73.43	91.63	74.16	53.86	88.82	55.38	93.35	62.05	77.22	83.00	92.02
DI-Net [39]	64.34	77.20	80.66	40.50	61.87	**87.06**	83.27	80.50	**92.45**	74.04	93.01	75.54	52.40	89.46	**58.37**	87.61	74.27	69.81	83.39	90.58
FE-Net [29]	65.31	77.90	80.87	47.83	67.86	87.04	82.82	77.26	88.23	77.77	94.45	74.28	49.35	89.88	48.50	93.14	74.81	76.79	83.43	91.91
Unet [17]	61.17	73.79	78.88	34.31	59.21	85.74	**87.62**	84.53	91.21	54.38	94.99	74.61	16.02	86.93	58.32	88.84	68.79	80.11	82.10	87.74
Unet++ [20]	61.39	74.41	79.37	26.82	61.84	81.50	83.68	84.33	89.08	73.70	92.10	74.70	29.26	89.67	54.29	85.35	76.58	75.78	82.33	91.51
Unet3+ [21]	61.29	74.87	77.25	27.56	62.46	67.88	84.29	83.85	84.12	75.54	88.11	70.16	61.13	87.41	55.85	86.48	76.97	72.43	82.64	91.04
MACU [36]	63.99	76.89	80.25	46.54	63.54	85.21	82.99	79.37	89.11	67.63	**94.57**	**76.67**	48.93	88.37	55.69	87.04	**80.20**	77.09	82.86	87.48
SBANet [49]	64.83	77.64	80.00	62.38	64.72	84.55	84.50	71.53	91.30	71.68	93.87	76.31	48.65	88.98	52.10	91.77	77.16	73.73	80.34	89.04
BAMTL [48]	65.55	78.31	80.82	53.63	63.37	86.89	81.36	80.40	91.72	79.23	95.32	76.68	**65.15**	88.32	53.71	90.81	74.57	79.83	83.38	87.64
FSCNet (our)	**67.31**	**79.62**	**81.64**	**55.08**	**66.32**	86.26	84.07	**88.88**	89.68	79.20	92.59	74.88	60.75	**90.32**	58.06	95.50	68.60	77.14	81.40	**93.04**

(*0: airplane, 1: bare-soil, 2: buildings, 3: cars, 4: chaparral, 5: court, 6: docks, 7: field, 8: grass, 9: mobile-home, 10: pavement, 11: sand, 12: sea, 13: ship, 14: tanks, 15: trees and 16: water*)

widely used networks, such as the FCN [14], Deeplabv3+ [15], GCN [18], PSPNet [19], Unet [17], Unet++ [20], and Unet3+ [21], all exhibited poor performances because they could not provide a more reasonable semantic feature description to model the coupled pixelwise relations regarding 17 categories of urban uses. In addition, the recently proposed urban use classification networks, CANet [35], MACU [36], DDCM [24], FE-Net [29], DI-Net [39], BAMTL [48], and SBANet [49], underperformed the proposed FSCNet, which was due to the noise from the shallow feature layer and naive semantic description existing in general CNNs that affect the performance of the fused semantic feature. Finally, several visualization results of the FCN [14], PSPNet [19], Unet3+ [21], MACU [36], FE-Net [29], and the proposed FSCNet are shown in Fig. 10. From the visualization results, the proposed FSCNet can provide more refined urban use classification results for both simple and complex semantic scenarios.

F. Ablation Study

As mentioned above, three datasets were used to demonstrate that the proposed FSCNet had a strong pixelwise semantic feature generalization ability, which could elegantly balance the pixelwise classification and localization performance in semantic segmentation by full semantic feature construction. In the following, we discuss the effectiveness of the designed modules in FSCNet.

1) Semantic Modalities Discussion: The effectiveness of each constructed full semantic modality in the proposed FSCNet is proven. The ablation study was conducted on ISPRS Vaihingen [51]. As shown in Table IV, the proposed FSCNet was divided into two parts: the encoder–decoder and semantic feature fusion. The different semantic modalities and their combinations were individually evaluated. In line 1 of Table IV, only the shallow semantic modality for final semantic prediction could achieve good performance with an 89.96% OA and an 80.86% mIoU caused by the optimized Unet3+ encoder–decoder strategy and reasonable interscale feature fusion of the designed ISF module. Next, compared with lines 2 and 3 of Table IV, the middle semantic modality outperformed the shallow and deep semantic modalities because SCC combined with ASPP can rectify the semantic feature and enable good classification and localization ability. Finally, in lines 4 and 5 of Table IV, the full semantic description, which contained the shallow, middle, and deep semantic modalities, outperformed the general shallow and deep layer (i.e., the low and high levels) semantic feature fusion methods. Therefore, from Table IV, the proposed full semantic construction strategy could effectively boost the pixelwise semantic feature description and balance the classification and localization in semantic segmentation.

2) Parameters and Inference Time Discussion: In addition, we evaluated the parameters and inference time of the proposed FSCNet with different backbones, as shown in Table V. Here, we selected four images from the ISPRS dataset to evaluate the inference time of handling one input image. From Table V, when the backbone was ResNet-18 [62], the proposed FSCNet obtained a good performance, with a 90.24% OA, an 89.26% mF1, and an 80.86% mIoU with a short inference time and few parameters. When the backbone was replaced by powerful EfficientNet-b6 [63], the proposed FSCNet obtained the SOTA performance, with a 91.11% OA, a 90.42% mF1, and an 82.75% mIoU, with acceptable parameters and a longer inference time. Therefore, the proposed FSCNet can be adaptively configured with different backbones to meet the processing requirements.

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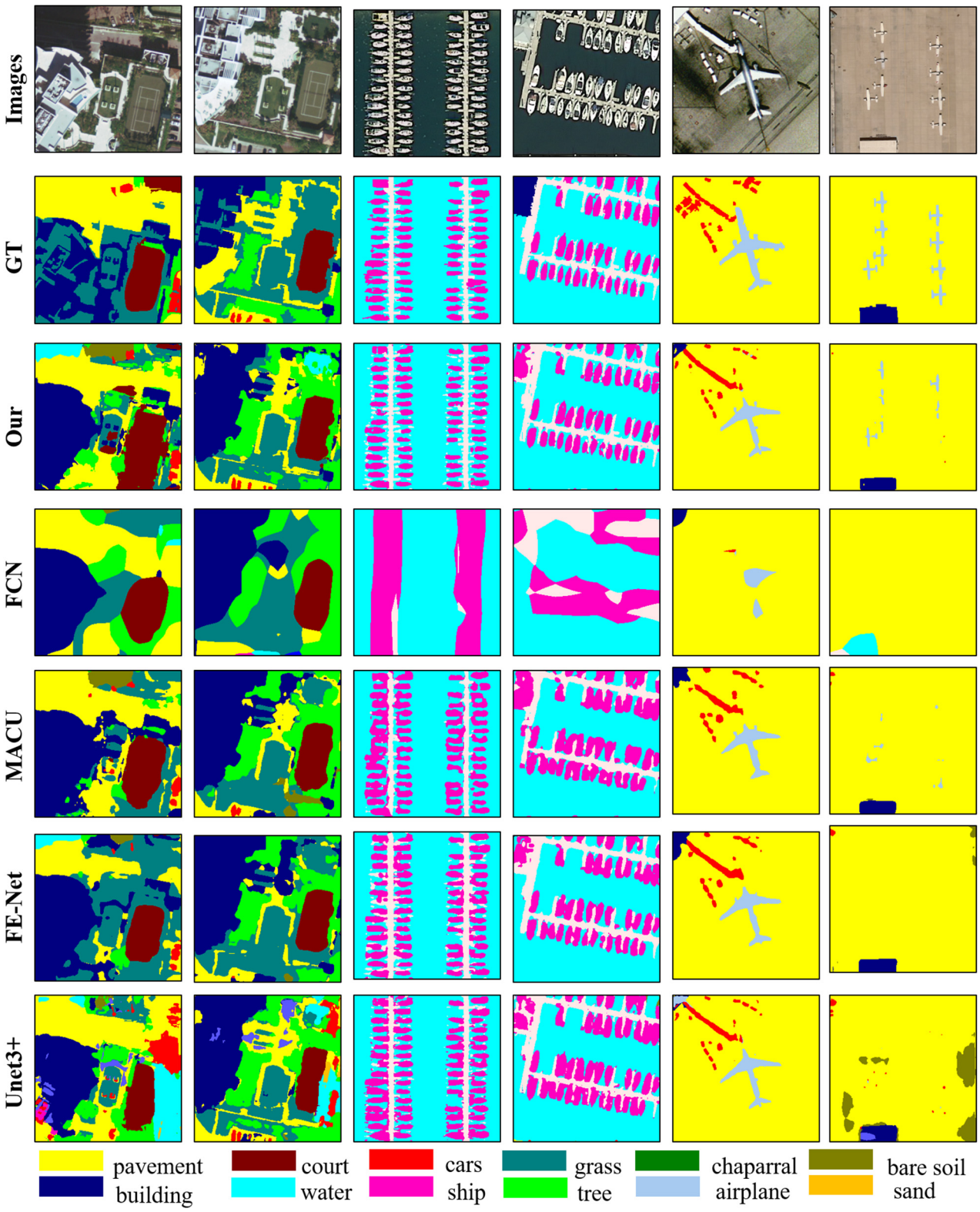


Fig. 10. Several visualized segmentation results of comparisons on the DLRSD [52] dataset.

3) Boundary Prediction Performance Discussion: Except for the robust pixelwise semantic feature generalization ability of the constructed full semantic description, the shallow semantic modality in the full semantic description was considered to support the boundary information extraction. By using an optimized Unet3+ [21] framework and then the designed ISF and MLCSA modules combined with boundary supervision in multitask loss, the shallow semantic modal-

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TABLE IV
ABLATION STUDY ON ISPRS VAIHINGEN

Encoder-Decoder		Semantic Feature Fusion			*OA(%)*	*mIoU(%)*
		#Shallow Layer	#Middle Layer	#Deep Layer		
Optimized Unet3+	ISF module	MLCSA	SCC+ASPP	PSP		
✓	✓	✓			89.96	80.86
✓	✓		✓		90.34	81.30
✓	✓			✓	89.93	79.86
✓	✓	✓		✓	90.32	81.16
✓	✓	✓	✓	✓	90.53	81.83

TABLE V
DIFFERENT BACKBONES TESTING ON ISPRS VAIHINGEN

Method/backbone		*OA (%)*	*mF1 (%)*	*mIoU (%)*	*Params *10^6*	*Times (s)*	*FLOPs (G)*
Unet [17]	ResNet-101 [62]	90.00	89.15	80.68	45.84	21	44.60
Unet++[20]		90.15	89.30	89.91	46.82	24	59.37
Unet3+[21]		90.08	89.45	81.16	55.63	47	312.64
MACUNet [36]		90.28	89.55	81.31	55.34	29	42.69
SBANet [49]		90.20	89.51	81.25	77.64	21	58.99
BAMTL [48]		90.13	89.35	81.00	76.42	22	60.57
FSCNet (ours)	ResNet-18 [62]	90.24	89.26	80.86	**15.68**	**21**	**45.59**
	ResNet-50 [62]	90.33	89.53	81.25	29.06	25	57.39
	ResNet-101 [62]	90.53	89.88	81.83	48.05	30	75.45
	ResNeSt-101 [63]	90.71	90.15	82.31	76.74	40	88.24
	EfficientNet-b6 [64]	**91.11**	**90.42**	**82.75**	84.65	48	52.84

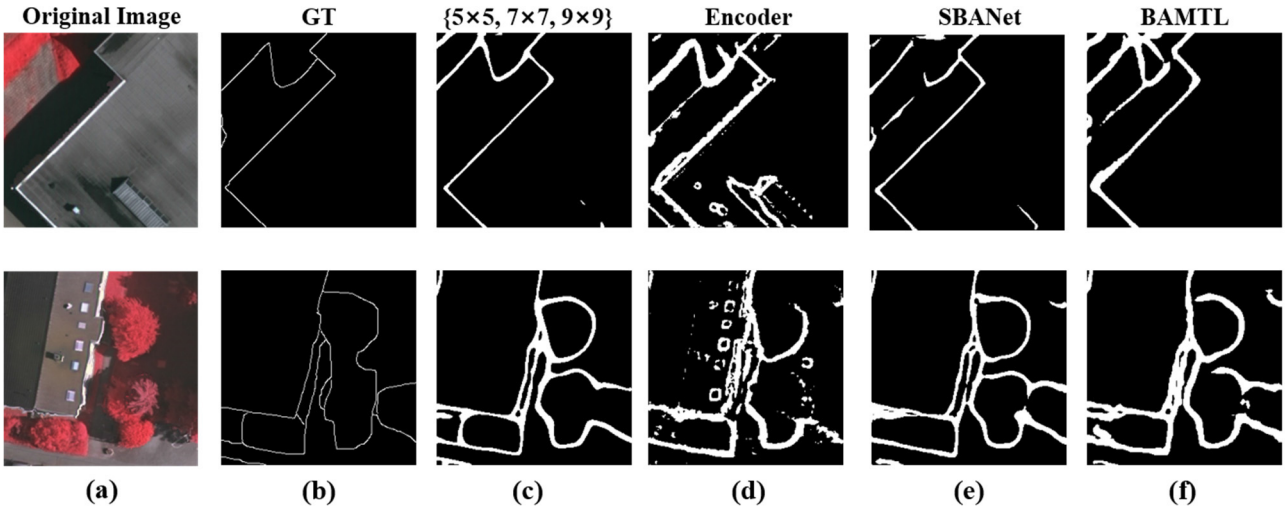


Fig. 11. Refined boundary prediction comparisons. (a) Original image. (b) GT. (c) Designed MLCSA. (d) Encoder. (e) SBANet [49]. (f) BANTL [48].

ity can capture the pure and valid boundary information and even achieve the refined boundary prediction. Moreover, we selected BAMTL [48] and SBANet [49], which also introduced boundary information to further improve the semantic feature representation. To prominently exhibit the boundary prediction ability of the proposed FSCNet, first, as shown in Table VI, the $\{5 \times 5, 7 \times 7, 9 \times 9\}$ multiscale context information adopted for the designed MLCSA module could facilitate the initial shallow semantic modality by eliminating random noise or interferences, and it could obtain the best boundary prediction, with a 91.18% OA and a 14.93% kappa. In addition, the same level features of the encoder and the decoder without the MLCSA module were also evaluated, and the results indicated that the larger multiscale context could

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TABLE VI
COMPARISONS OF BOUNDARY PREDICTION ON ISPRS VAIHINGEN

Thres	Index	Encoder	NoAtten	3×3	{3×3, 5×5}	{5×5, 7×7, 9×9}	SBANet [49]	BAMTL [48]
0.9	*OA*	90.44	90.64	90.71	90.63	91.18	90.74	87.50
	Kappa	12.16	15.29	15.38	15.28	15.46	14.93	11.96
0.6	*OA*	86.45	89.50	89.70	89.58	90.16	89.72	85.47
	Kappa	10.60	14.63	14.79	14.68	14.88	14.35	11.20
0.3	*OA*	81.65	88.43	88.72	88.59	89.19	88.75	83.58
	Kappa	8.97	14.05	14.23	14.13	14.34	13.82	10.54

(Note: **Thres** represents threshold; **Encoder** means the shallow layer feature from backbone; **NoAtten** indicates the initial shallow semantic modality feature from optimized Unet3+ and ISF module)

effectively eliminate random noise in the shallow semantic modality by spatial attention weights. Next, as reported in Table VI, BAMTL [48] and SBANet [49] both designed the boundary extraction module applied for deep semantic feature description. However, deep layer features had poor localization ability; therefore, they could not provide the refined boundary information, which would lead BAMTL [48] and SBANet [49] to provide bad results. Finally, several visualized boundary prediction results are shown in Fig. 11. In Fig. 11(c), for $\{5 \times 5, 7 \times 7, 9 \times 9\}$ multiscale context used for the designed MLCSA module, the random noise or interference in the encoding shallow layer features was removed, which was beneficial for fine-scale boundary prediction and provides the more accurate boundary prediction results than other methods.

V. CONCLUSION

In this article, a novel full semantic construction network called FSCNet is proposed for refined pixelwise classification from VHR complex urban scenes. First, we analyze the ERF changes in stacked convolution layers to find that unreasonable ERF changes would lead to the formation of a naive semantic description, which generates more inaccurate localization and classification results for pixelwise urban use classification. Second, to improve the naive semantic description for refined urban use classification, shallow, middle, and deep semantic modalities are rebuilt to redefine full semantic construction. Then, to consider a large number of small-scale ERFs that exist in the naive semantic description, MLCSA and boundary supervision are designed to leverage the shallow semantic modality to capture the basic and valid boundary information. Next, related to fewer ERFs that are not large enough in the naive semantic description, the PSP module is employed for the deep semantic modality to further enlarge the receptive field and capture more abstract global information. Furthermore, according to a severe semantic gap between redefined shallow and deep semantic modalities, the middle semantic modality is also redefined and rebuilt by the designed SCC combined with ASPP, which can bridge the semantic gap and is beneficial for full semantic construction. Finally, extensive experiments were carried out on three published urban use classification datasets, and the results showed that the proposed FSCNet can not only provide refined pixelwise urban use classification but also outperform other SOTA networks. Then, for further work, we will continue to explore the decoupling method of semantic feature extraction in implicit CNN architectures.

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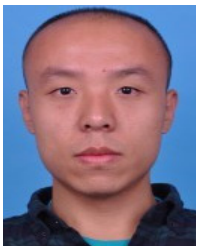
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